```
000001          IDENTIFICATION DIVISION.
000002          PROGRAM-ID. MatrixCalculation.
000003
000004          ENVIRONMENT DIVISION.
000005          DATA DIVISION.
000006
000007          WORKING-STORAGE SECTION.
000008
000009          01  M-SIZE          PIC 9(4) COMP VALUE 400.
000010
000011          01  RESULT1         PIC S9(14) COMP VALUE 0.
000012          01  RESULT2         PIC S9(14) COMP VALUE 0.
000013
000014          01  TIME1           COMP-2 VALUE 0.
000015          01  TIME2           COMP-2 VALUE 0.
000016
000017          01  M1.
000018              05  M1-ROW OCCURS 400 TIMES.
000019                  10  M1-ELEMENTS OCCURS 400 TIMES PIC S9(10) COMP.
000020          01  M2.
000021              05  M2-ROW OCCURS 400 TIMES.
000022                  10  M2-ELEMENTS OCCURS 400 TIMES PIC S9(10) COMP.
000023          01  M3.
000024              05  M3-ROW OCCURS 400 TIMES.
000025                  10  M3-ELEMENTS OCCURS 400 TIMES PIC S9(10) COMP.
000026
000027          01  I               PIC S9(10) COMP.
000028          01  J               PIC S9(10) COMP.
000029          01  K               PIC S9(10) COMP.
000030
000031          01  START-TIME      PIC S9(10) COMP.
000032          01  END-TIME        PIC S9(10) COMP.
000033
000034          01  TEMP            PIC S9(10) COMP.
000035
000036          01  RESULT1_TXT     PIC 9(14).
000037          01  RESULT2_TXT     PIC 9(14).
000038          01  TIME1_TXT       PIC 9(3).9(6).
000039          01  TIME2_TXT       PIC 9(3).9(6).
000040
000041          PROCEDURE DIVISION.
000042
000043        * ------------------------------------------------
000044
000045              ACCEPT START-TIME FROM TIME.
000046
000047              MOVE 0 TO RESULT1.
000048
000049              PERFORM VARYING I FROM 1 BY 1 UNTIL 100 < I
000050                  PERFORM VARYING J FROM 1 BY 1 UNTIL 1000000 < J
000051                      ADD J TO RESULT1
000052                  END-PERFORM
000053              END-PERFORM.
000054
000055              ACCEPT END-TIME FROM TIME.
000056              COMPUTE TIME1 = (END-TIME - START-TIME) / 100.
000057
000058        * ------------------------------------------------
000059
000060              ACCEPT START-TIME FROM TIME.
000061
000062              PERFORM VARYING I FROM 1 BY 1 UNTIL M-SIZE < I
000063                  PERFORM VARYING J FROM 1 BY 1 UNTIL M-SIZE < J
000064                      ADD I TO J GIVING M1-ELEMENTS(I, J)
000065                      ADD I TO J GIVING M2-ELEMENTS(I, J)
000066                  END-PERFORM
000067              END-PERFORM.
000068
000069              PERFORM VARYING I FROM 1 BY 1 UNTIL M-SIZE < I
000070                  PERFORM VARYING J FROM 1 BY 1 UNTIL M-SIZE < J
000071                      MOVE 0 TO M3-ELEMENTS(I, J)
000072                      PERFORM VARYING K FROM 1 BY 1 UNTIL M-SIZE < K
000073                          MULTIPLY M1-ELEMENTS(I, K) BY M2-ELEMENTS(K, J)
000074                              GIVING TEMP
000075                          ADD TEMP TO M3-ELEMENTS(I, J)
000076                      END-PERFORM
000077                  END-PERFORM
000078              END-PERFORM.
000079
000080              MOVE 0 TO RESULT2.
```

```
000081
000082                PERFORM VARYING I FROM 1 BY 1 UNTIL M-SIZE < I
000083                    PERFORM VARYING J FROM 1 BY 1 UNTIL M-SIZE < J
000084                        ADD M3-ELEMENTS(I, J) TO RESULT2
000085                    END-PERFORM
000086                END-PERFORM.
000087
000088                ACCEPT END-TIME FROM TIME.
000089                COMPUTE TIME2 = (END-TIME - START-TIME) / 100.
000090
000091              * ---------------------------------------------------
000092
000093                MOVE RESULT1 TO RESULT1_TXT
000094                MOVE RESULT2 TO RESULT2_TXT
000095                MOVE TIME1   TO TIME1_TXT
000096                MOVE TIME2   TO TIME2_TXT
000097
000098                DISPLAY 'COBOL   : Result= ' RESULT1_TXT
000099                        ', Time= ' TIME1_TXT ' sec'
000100                        ', Result= ' RESULT2_TXT
000101                        ', Time= ' TIME2_TXT ' sec'.
000102
000103             STOP RUN.
000104
```